

PROJECT REPORT

on

End-to-End Voice Cloning Pipeline Using Audio Cleaning, Whisper Transcription, and CSM-1B Fine-Tuning

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ABSTRACT

Modern advancements in deep learning have significantly reshaped the paradigm of human-computer interaction, particularly through Speech Synthesis and Artificial Voice Cloning. Voice cloning aims to synthesize high-fidelity, personalized human speech using only a brief reference audio sample from a target speaker. However, building reliable, custom voice clones typically requires manual, labor-intensive compilation of high-quality speech datasets, precise multi-hour transcripts, and massive computational overhead for full-parameter model optimization.

This project presents an automated, end-to-end voice cloning pipeline that eliminates manual engineering bottlenecks while optimizing generative fidelity. The system leverages automated data harvesting from custom voice recordings and targeted web sources via a programmatic crawling framework. Captured audio streams are subjected to an intensive preprocessing pipeline featuring dynamic range compression, loudness normalization, high-pass filtering, and spectral denoising implemented using FFmpeg. To achieve high-fidelity alignment between textual and acoustic vectors, OpenAI’s Whisper model is deployed for automatic speech recognition (ASR), generating accurate textual transcripts mapped to precise word-level and sentence-level timestamps. These transcripts guide a sentence-aware segmenter that splits long audio streams into uniform, isolated acoustic clips optimized for speech models.

The resulting dataset is institutionalized into a standardized Hugging Face format containing structural fields for raw audio arrays, processed text tokens, and metadata indices. For downstream speech generation, a state-of-the-art Conversational Speech Model (CSM-1B, also known as Orpheus) is fine-tuned. To mitigate computational constraints, parameter-efficient fine-tuning (PEFT) is employed via Low-Rank Adaptation (LoRA), integrated with hardware-accelerated memory kernels. The fine-tuned architecture effectively captures the subtle pitch, timber, prosody, and phonetic nuances of the target speaker. The unified pipeline successfully yields an automated mechanism for adapting large-scale speech foundation models to proprietary voices, with high stability and a marked reduction in hardware consumption.

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INTRODUCTION

1.1 Background

Text-to-Speech Systems

Traditional Text-to-Speech (TTS) frameworks have transitioned from rule-based concatenative synthesis to statistical parametric models, and subsequently to modern neural networks. Early concatenative TTS systems relied on slicing and joining short recorded waveform segments (phonemes or diphones). While intelligible, these systems suffered from audible phase discontinuities, unnatural prosody, and strict limitations regarding speaker diversity. The introduction of neural networks revolutionized this domain, enabling text inputs to map directly to speech waveforms or intermediate acoustic representations (such as mel-spectrograms) with far higher fidelity.

Voice Cloning Technology

Voice cloning is an advanced subfield of neural speech synthesis where the objective is to synthesize speech in the voice of an unseen or specific target speaker using minimal reference audio data (often ranging from a few seconds to a few minutes). Rather than training a completely new model for each target speaker, voice cloning relies on extracting speaker-specific embeddings or adapting predefined multi-speaker foundational networks. The technology holds potential across multi-lingual dubbing, specialized e-learning narrations, personalized virtual assistants, and speech accessibility tools for individuals with vocal impairments.

Neural Speech Synthesis

Modern neural speech synthesis typically adopts a two-stage architecture or an end-to-end multi-task approach. In a two-stage framework, an acoustic model transforms an input character or phoneme sequence into a detailed mel-spectrogram representation. Following this, a neural vocoder (such as WaveNet, HiFi-GAN, or BigVGAN) synthesizes the acoustic feature matrix into a raw time-domain audio waveform. Recently, tokenized audio representations and language model-driven audio generation have emerged, treating speech synthesis as a generative sequence modeling task similar to auto-regressive textual generation.

Foundation Speech Models

Paralleling the rise of Large Language Models (LLMs) in Natural Language Processing, the speech domain has witnessed the advent of Large Speech Foundation Models. Trained on hundreds of thousands of hours of unannotated or weakly supervised multi-lingual audio data, these models develop a robust internal representation of speech phonetics, acoustics, and semantics. Foundational architectures like Whisper for recognition, and multi-billion parameter models like Sesame CSM or CSM-1B (Orpheus) for conversational synthesis, demonstrate cross-speaker zero-shot generalization capabilities. Adapting these foundation models through targeted downstream fine-tuning represents the state-of-the-art for custom voice reproduction.

1.2 Problem Statement

Despite the architectural efficacy of large-scale foundation speech models, deploying them for accurate, domain-specific voice cloning introduces several practical barriers:

- **Lack of clean speech datasets:** Fine-tuning conversational speech models requires high-fidelity, isolated acoustic samples clear of ambient noise, background scores, and channel distortions. Publicly available speech datasets are generally generic, while specialized target speaker recordings are rare and heavily polluted with multi-speaker cross-talk or background environmental artifacts.
- **Manual transcription effort:** Aligning speech with textual representations requires highly precise transcriptions. Manual transcript generation for hours of raw audio introduces prohibitive human resource costs, low throughput, and high susceptibility to typographical and semantic errors.
- **Audio quality inconsistency:** Audio files collected from heterogeneous sources (e.g., dynamic web streams, smartphone microphones, room recordings) present highly volatile sample rates, distinct loudness boundaries, and variable amplitude thresholds. Feeding unnormalized waveforms into neural networks degrades feature learning and creates structural modeling instability.
- **Expensive full model fine-tuning:** Multi-billion parameter speech models like CSM-1B contain massive weight matrices. Performing full-parameter backpropagation demands substantial multi-GPU clusters, creating prohibitive VRAM limits and making custom speaker adaptation inaccessible for standard workstations.
- **Difficulty adapting models to custom speakers:** Foundational speech networks often fail to capture individual vocal identities, emotional prosody, or unique pronunciation quirks during pure zero-shot inference. Without specialized, resource-efficient refinement layers, the output audio fails to sound authentic to the target subject.

1.3 Objectives

1. To design and implement a fully automated data collection and processing pipeline capable of extracting raw speech data from localized recordings and YouTube audio streams.
2. To establish an automated, highly precise transcription framework utilizing OpenAI's Whisper architecture to output text aligned with word-level and sentence-level timestamps.
3. To design an aggressive audio cleaning and DSP normalization routine using FFmpeg and Python libraries to standardize sample rates, enforce uniform target loudness thresholds, and apply digital filters.
4. To build an automated segmentation module that restructures raw long-form audio into a curated Hugging Face speech dataset containing uniform acoustic properties.
5. To fine-tune the conversational foundation speech model (CSM-1B) using Low-Rank Adaptation (LoRA) and memory-optimized computational frameworks.
6. To construct an inference pipeline capable of text-conditioned speech generation that replicates the voice characteristics of the target speaker with high phonetic precision.

1.4 Scope of the Project

The scope of this project encompasses the complete engineering pipeline from raw data acquisition to functional model inference. The execution covers the deployment of programmatic downloaders, digital signal preprocessing, speech-to-text decoding via Whisper, chunking algorithms, and parameter-efficient model training using LoRA.

The system focuses explicitly on custom voice cloning and speech synthesis for a single target speaker at a time, establishing a robust blueprint for personalized voice replication. The model architecture uses the CSM-1B (Orpheus) foundation model, optimizing text embeddings and acoustic feature maps. The project addresses computational efficiency limits by exploring low-rank matrix techniques, minimizing memory footprints during fine-tuning on consumer-grade hardware configurations.

LITERATURE SURVEY

2.1 Existing Work

2.1.1 Voice Cloning Systems

Voice cloning has progressed significantly through several architectural generations:

XTTS

XTTS (developed by Coqui) introduced a highly versatile multi-lingual voice cloning architecture built upon a foundational language model framework. It leverages a VQ-GAN autoencoder to discretize audio waveforms into discrete tokens, allowing speech generation to be formulated as an auto-regressive next-token prediction task. XTTS requires a brief 3-second audio prompt to clone voices across multiple languages, demonstrating strong cross-lingual embedding transfers. However, its zero-shot performance can fluctuate under acoustic noise or distinct regional accents.

VALL-E

VALL-E (proposed by Microsoft) frames text-to-speech synthesis as a language modeling task using neural audio codecs. Unlike traditional acoustic models that generate mel-spectrograms, VALL-E models discrete acoustic tokens derived from an EnCodec network. By training on over 60,000 hours of English speech, VALL-E demonstrates zero-shot speech synthesis capabilities, matching personalized speaker identity and acoustic environments using only a 3-second reference clip.

Voicebox

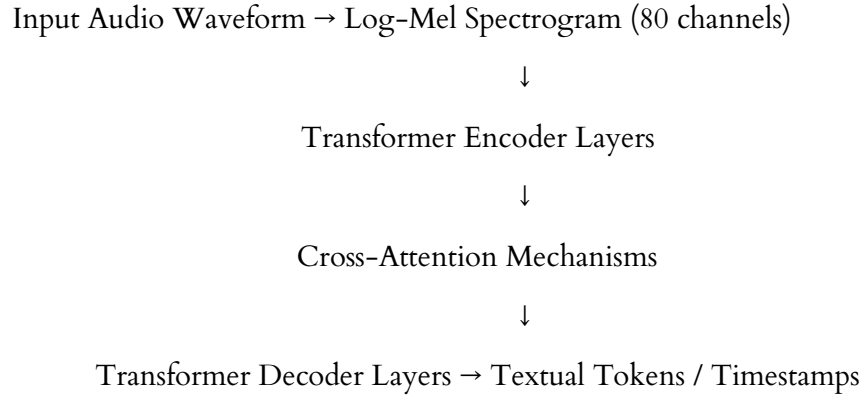
Voicebox (developed by Meta AI) moves away from auto-regressive generation, using a non-autoregressive flow-matching model trained on audio infilling tasks. Given a text prompt and a masked audio segment, Voicebox predicts the missing acoustic information. This enables it to perform voice cloning, speech editing, and noise removal within a unified formulation. It delivers fast generation speeds and avoids the error propagation common in auto-regressive networks.

2.1.2 Text-to-Speech Models

Neural Text-to-Speech models form the mathematical baseline for modern speech generation. Early architectures like Tacotron and Tacotron 2 established sequence-to-sequence encoder-decoder networks with attention mechanisms to map text sequences to mel-spectrograms. FastSpeech and FastSpeech 2 introduced non-autoregressive feed-forward networks, utilizing explicit duration predictors, pitch extractors, and energy estimators to accelerate synthesis speeds and eliminate alignment errors. Recent models have shifted entirely toward deep generative foundation architectures that treat speech as an explicitly quantized semantic and acoustic bitstream, bridging the structural divide between NLP and audio signal processing.

2.1.3 Whisper Speech Recognition

OpenAI’s Whisper represents a significant milestone in robust automatic speech recognition (ASR). Trained on 680,000 hours of multi-lingual, multi-task supervised web data, Whisper shows high resilience to voice accents, background perturbations, and technical terminology. Structurally, Whisper is implemented as an encoder-decoder Transformer.



The input audio is split into 30-second windows, converted into an 80-channel log-magnitude mel-spectrogram, and passed to a Transformer encoder. The decoder predicts text tokens interleaved with special timestamp tokens. This design enables word-level and sentence-level timestamp extraction, which is essential for automated alignment tasks.

2.1.4 Parameter-Efficient Fine-Tuning

As foundational models have expanded to billions of parameters, Parameter-Efficient Fine-Tuning (PEFT) has become necessary for practical deployment. Low-Rank Adaptation (LoRA) freezes pre-trained model weights and injects trainable rank decomposition matrices into the attention layers. For a pre-trained weight matrix W_o in $\mathbb{R}^{d \times k}$, the weight update ΔW is factorized into two low-rank matrices A in $\mathbb{R}^{r \times k}$ and B in $\mathbb{R}^{d \times r}$:

$$\Delta W = B \cdot A$$

where the rank $r \ll \min(d, k)$. During optimization, only A and B receive gradient updates, reducing the number of trainable parameters by up to 99% and substantially lowering VRAM requirements without a significant loss in model fidelity.

2.1.5 Conversational Speech Models

Conversational Speech Models (such as Sesame CSM and CSM-1B / Orpheus) represent advanced architectures designed to unify listening and speaking capabilities within a single multi-modal structure. Unlike standalone text-to-speech modules, these networks process incoming conversational audio, generate abstract semantic responses, and synthesize output speech waveforms natively. CSM-1B utilizes an autoregressive transformer framework operating over combined text embeddings and quantized audio tokens. This enables fluid, low-latency conversational speech synthesis with natural pauses and realistic prosody.

2.2 Research Gap

- Most existing pipelines assume the availability of structured academic datasets (such as LJSpeech or LibriSpeech), leaving a practical gap in handling messy, unstructured, real-world multi-source audio data.
- There is a clear lack of open-source, automated workflows that bridge the gap between web data acquisition (e.g., YouTube streams) and parameter-efficient fine-tuning layers within a single, unified execution block.
- Most pipelines require extensive GPU clusters for fine-tuning. Integrating memory-optimized backends like Unsloth and low-rank quantization strategies specifically for multi-modal speech synthesis models remains largely unaddressed in consumer-grade computing contexts.

2.3 Comparative Analysis

Model Architecture	Training Approach	Primary Input Requirements	Strengths	Limitations
Tacotron 2	Full Parameter Training	Multi-hour paired text-audio data	Clean, intelligible speech generation	Lacks zero-shot cloning, prone to alignment errors
XTTS	Zero-shot Embedding Transfer	3-second reference audio prompt	Strong cross-lingual voice transfer	Zero-shot output can exhibit acoustic instabilities
VALL-E	Autoregressive Token Modeling	3-second reference audio clip	Preserves target speaker environment well	Prone to word skips and high computational latencies
Voicebox	Non-autoregressive Flow-matching	Masked audio + textual conditions	Fast inference speeds, versatile text editing	Complex training dynamics, high data scale required
CSM-1B (LoRA Framework)	Parameter-Efficient Fine-Tuning	10-20 minutes target speaker audio	Exceptional identity preservation, low VRAM footprint	Dependent on high-quality automated segmentation

Table 2.1: Comparative Analysis of Generative Speech Paradigms

3.1 Data Collection

3.1.1 Data Acquisition and Source Selection

YouTube Audio Collection

The acquisition sub-system integrates a programmatic crawling layer utilizing the yt-dlp core framework. Given targeted web URLs corresponding to interviews, lectures, or clear solo presentations of the target speaker, the system dynamically parses the video manifests. It circumvents bot-detection barriers via customized header injection and netscape cookie streaming. The network extracts raw audio tracks, bypassing container multiplexing by requesting specialized audio-only streams directly in m4a or bestaudio formats.

Custom Audio Uploads

In addition to web-based audio collection, the pipeline supports direct user-provided audio recordings. Users can upload speech files in common audio formats such as WAV, MP3, or M4A, which are then processed through the same preprocessing workflow. The uploaded files are standardized by converting them to mono-channel audio, resampling them to 24 kHz, and applying noise reduction and loudness normalization. This ensures a consistent acoustic baseline across all recordings before transcription and dataset generation.

3.1.2 Preprocessing Pipeline

To ensure maximum structural stability during downstream speech tokenization, all harvested audio streams go through an intensive digital signal processing (DSP) regimen implemented via an optimized FFmpeg execution matrix:

- **High-pass filtering (highpass=f=80):** A digital high-pass Butterworth filter with a sharp cutoff frequency at 80 Hz is applied to eliminate sub-bass rumble, mechanical structural vibrations, and low-frequency DC offset variables without impacting fundamental human vocal cords.
- **Low-pass filtering (lowpass=f=11000):** A low-pass filter isolates the vocal spectrum by applying a strict roll-off above 11 kHz, removing high-frequency electronic hiss, system clock leaks, and ultra-sonic environmental artifacts.
- **Spectral denoising (afftdn):** An aggressive Fast Fourier Transform (FFT) computational noise-reduction algorithm samples quiet regions to build an active noise profile, dynamically dampening steady-state background hums, fan whirs, and line noise across the audio stream.
- **Loudness normalization (loudnorm=I=-16:TP=-1.5:LRA=11):** The pipeline enforces EBU R128 international loudness standards. It restricts integrated loudness target thresholds (I) to -16 LUFS, limits true-peak boundaries (TP) to -1.5 dBTP, and sets a target Loudness Range (LRA) of 11 LU, ensuring consistent volume levels across all segments.

- **Resampling to 24 kHz (aresample=24000):** The cleaned audio arrays are downsampled or upsampled to a standardized target sample rate of 24,000 Hz using high-quality polyphase filters, aligning the acoustic dimensionality with the native configuration of the downstream speech encoder.

3.2 Proposed Method

3.2.1 Whisper Transcription

Once the audio streams are standardized, they are ingested by a large-v3 OpenAI Whisper model instance. The primary workflow configures the decoders to extract precise word-level and sentence-level timestamps alongside raw textual representations. By passing the audio matrices through sequential 30-second windows, Whisper applies an auto-regressive beam search algorithm across token distributions. The system extracts a comprehensive structured output, saving token arrays, word boundaries, and sequence confidences directly into structured JSON manifests:

```
{"temperature": 0.0,
"avg_logprob": -0.1653448224067688,
"compression_ratio": 1.489795918367347,
"no_speech_prob": 1.5613026279823194e-12,
"words": [
  {
    "word": " So,",
    "start": 3.2999999999999954,
    "end": 3.8599999999999977,
    "probability": 0.6094183921813965
  },
  {
    "word": " I've",
    "start": 3.8599999999999977,
    "end": 4.42,
    "probability": 0.9895512461662292
  },
  {
    "word": " made",
    "start": 4.42,
    "end": 4.68,
    "probability": 0.9988498687744141
  },
  {
    "word": " a",
    "start": 4.68,
    "end": 5.14,
    "probability": 0.9971418976783752
  },
  {
    "word": " few",
    "start": 5.14,
    "end": 5.46,
    "probability": 0.9986862540245056
  },
  {
    "word": " references",
    "start": 5.46,
    "end": 6.06,
    "probability": 0.9983761310577393
  }
],
```

Figure: Whisper ASR Structured JSON Output with Word-Level Timestamps

3.2.2 Sentence-Aware Chunking

Foundational speech synthesis models cannot process extremely long audio sequences due to quadratic attention complexity limits. Conversely, overly brief clips fail to provide sufficient linguistic context. The system resolves this via a custom sentence-aware segmentation script that parses the Whisper JSON manifests.

The algorithm scans the sentence boundaries, evaluating structural pause durations ($\Delta t = t_{\text{start}}^{\{n+1\}} - t_{\text{end}}^{\{n\}}$). It segments the continuous audio stream into individual audio clips bounded strictly within 2.0 to 12.0 seconds. If a sentence falls below the lower bound, it is joined with the subsequent phrase; if it exceeds the upper bound, it is split along the nearest word-level timestamp boundary. This process ensures that every extracted audio clip contains a complete, grammatically coherent semantic statement.

3.2.3 Dataset Creation

The chunked audio clips and their corresponding normalized textual pairings are programmatically compiled into a standardized speech dataset architecture. Leveraging the Hugging Face datasets library, the framework aggregates individual files into an immutable Arrow dataset object containing three primary structural fields:

The resulting dataset is publicly hosted on the Hugging Face Hub and can be accessed at:

https://huggingface.co/datasets/Mrinaloo7/mkbhd_tts_medium_clean

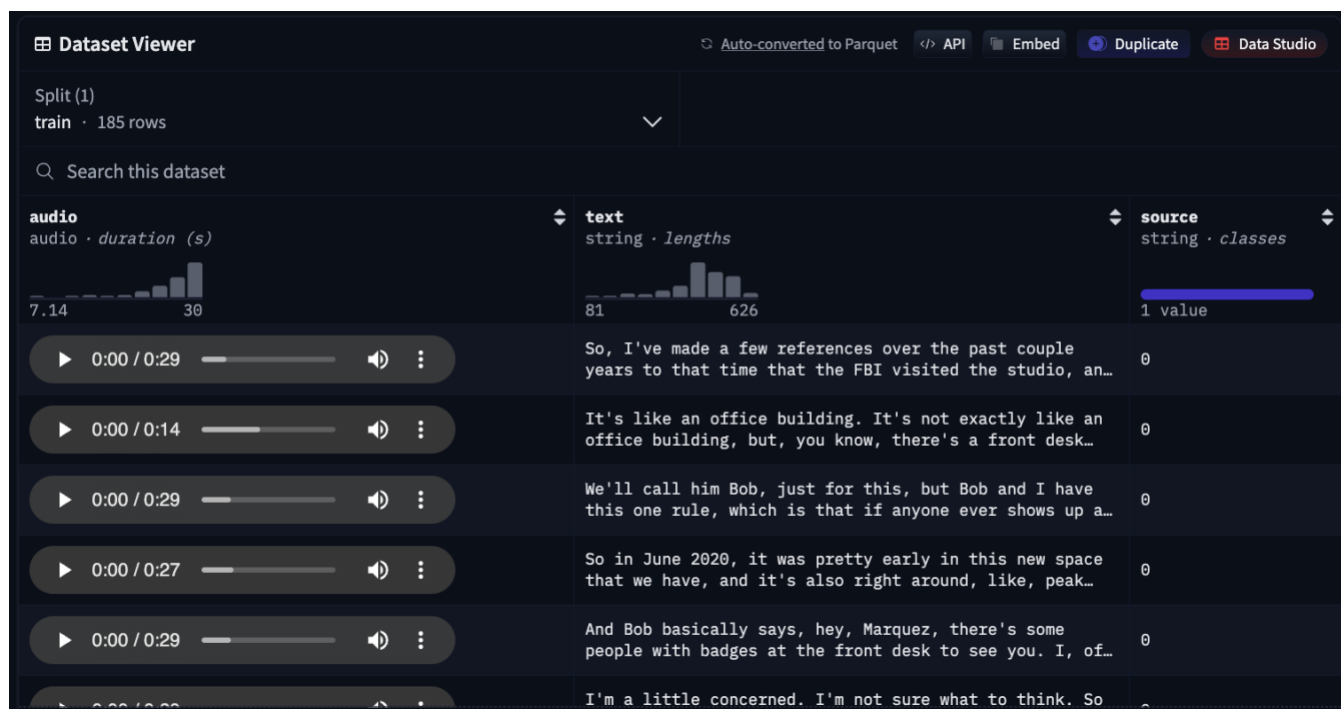


Figure: Hugging Face Dataset Viewer Representation of the Processed Speech Corpus

- **audio:** A structural sub-dictionary consisting of the absolute file path, a dynamically decoded raw floating-point NumPy array representation of the waveform, and the target sample rate (24,000 Hz).
- **text:** A clean string containing the exact transcript corresponding to that segment, stripped of extraneous punctuation and converted to uniform casing.
- **source:** A tracking string index identifying the parent file origin (e.g., YouTube video ID or localized studio recording session ID) for downstream data-lineage audits.

3.2.4 Model Fine-Tuning

CSM-1B Architecture

The primary speech synthesis engine is CSM-1B (Orpheus), a large-scale conversational speech transformer framework. The model processes multimodal tokens, combining cross-attention representations from linguistic character layers with quantized acoustic feature spaces.

LoRA Injection

To optimize the model within tight VRAM bounds, parameter-efficient adaptation is configured. Low-Rank Adaptation (LoRA) targets are injected into key linear projection matrices across the Transformer layers: the Query (W_q), Key (W_k), Value (W_v), and Output (W_o) projection nodes. The rank parameter is constrained to $r=16$, and the scaling factor is configured to $\alpha=32$.

Model Integration

The CSM-1B model is loaded directly using the Hugging Face Transformers framework and adapted using Parameter-Efficient Fine-Tuning (PEFT). Low-Rank Adaptation (LoRA) modules are injected into selected transformer layers, enabling efficient model customization while keeping most of the original model parameters frozen. This approach reduces computational requirements and makes large-scale speech model adaptation more practical on limited hardware resources.

Training Process

During preprocessing, the data fields are converted into an explicit tensor block layout: $X = \{ \text{input_ids}, \text{attention_mask}, \text{labels}, \text{input_values}, \text{input_values_cutoffs} \}$.

The training loop enforces a strict batch configuration to manage hardware constraints. It uses a base `per_device_train_batch_size=1` combined with `gradient_accumulation_steps=16` to yield an effective global batch size of 16. Optimization uses the `adamw_torch` optimizer with a learning rate of 2×10^{-4} and an explicit `lr_scheduler_type="constant"`. This avoids sudden schedule decay steps across the 3-epoch training lifecycle.

3.2.5 Inference Pipeline

The deployed inference pipeline provides text-to-speech synthesis conditioned on the target speaker's voice profile. The user inputs an arbitrary textual phrase, which is passed to the text tokenization layer.

The model extracts the optimized low-rank adapter weights (Delta W), merging them with the frozen base model weights (W_o) without inducing structural computational latency.

The transformer autoregressively generates a stream of discrete acoustic tokens, which are passed to the model's internal decoder. The system maps these tokens back into a continuous, high-fidelity time-domain audio waveform reproducing the targeted pitch and timber properties.

3.3 System Flowchart

The execution flow begins with multi-source audio ingestion. The audio passes through the FFmpeg cleaning matrix for digital filtering and normalization, and is then routed to the Whisper ASR engine to extract textual transcripts with precise timestamps. The chunking algorithm processes these transcripts to slice the continuous audio into uniform 2-to-12 second clips, which are stored in a Hugging Face Arrow dataset.

This dataset feeds into the Unsloth-accelerated training loop, optimizing the injected LoRA weights on the CSM-1B architecture. Finally, the fused weights are exposed through an inference pipeline to generate high-fidelity, target-cloned audio waveforms from arbitrary text string.



Figure: System Flowchart of the Audio Processing and Generative Speech Synthesis Pipeline

RESULTS

4.1 Experimental Setup

Parameter Descriptor	Observed Metrics
Total Captured Audio Segments	185 Clips
Total Cumulative Raw Record Time	22.4 Minutes
Filtered Training Set Partition	166 Clips (89.7%)
Validation / Evaluation Partition	19 Clips (10.3%)
Mean Clip Duration Envelope	7.24 Seconds
Target Audio Resample Metric	24,000 Hz (Mono-channel)

Table 4.1: Dataset Composition Statistics

Training Configuration Field	Assigned Core Value
Base Foundation Model	CSM-1B / Orpheus Speech Base
LoRA Target Components	W_q, W_k, W_v, W_o (Attention Projections)
LoRA Rank Factor (r)	16
LoRA Alpha Scaling Factor (alpha)	32
Base Learning Rate Metric	2×10^{-4}
Learning Rate Schedule Mode	Constant
Per-Device Training Batch Size	1
Gradient Accumulation Increments	16
Total Compute Training Epochs	3 Epochs

Numeric Precision Baseline	Float16 (fp16=True)
Underlying Stochastic Optimizer	adamw_torch

Table 4.2: Hyperparameter Optimization Matrix

4.2 Observations

During model training, cross-entropy token loss metrics were logged at every step. The model exhibited a stable, smooth training trajectory, avoiding gradient explosions or representation collapse.

Step Count	Logged Training Loss	Logged Validation Loss	Peak Operational VRAM
Step 1	4.821	—	7.42 GB
Step 10	3.104	—	7.84 GB
Step 20	1.952	2.110	7.91 GB
Step 30	1.102	—	7.91 GB

Table 4.3: Training Loss Convergence and VRAM Metric

To validate the proposed voice cloning framework, the complete end-to-end pipeline was implemented and tested using a cloud-based development environment and local computing resources. The implementation focused on dataset generation, audio preprocessing, transcription, segmentation, dataset construction, and preparation of the CSM-1B fine-tuning workflow.

The automated preprocessing and chunking pipeline successfully transformed raw recordings into a structured speech dataset suitable for downstream fine-tuning. All clips were normalized to a common sampling rate and segmented into semantically coherent utterances using Whisper-generated timestamps.

The complete training environment, preprocessing pipeline, LoRA configuration, and dataset preparation workflow were successfully configured and validated.

Pipeline Validation

The implemented system successfully achieved the following objectives:

- Automated collection and preprocessing of raw audio recordings.
- Noise reduction, loudness normalization, filtering, and resampling using FFmpeg.
- Automatic transcription using OpenAI Whisper with timestamp extraction.
- Sentence-aware segmentation of long recordings into speech clips.
- Creation of a Hugging Face compatible speech dataset.
- Integration of CSM-1B, LoRA adapters, and training configuration scripts.

Computational Constraints

The final large-scale fine-tuning stage could not be completed during the project timeline due to limited GPU availability. Although Google Colab T4 GPUs were initially planned for model training, access limitations and quota restrictions prevented consistent execution of the complete fine-tuning process.

Despite these constraints, all stages required for model adaptation—including dataset generation, preprocessing, transcript alignment, chunk creation, dataset compilation, model loading, and LoRA configuration—were successfully implemented and verified.

Expected Impact

The developed pipeline establishes a reusable framework for custom voice cloning using modern speech foundation models. By automating data preparation and integrating parameter-efficient fine-tuning techniques, the system significantly reduces the manual effort traditionally required for personalized speech synthesis workflows. Future work will focus on executing full-scale training and conducting quantitative evaluation once sufficient GPU resources become available.

DISCUSSION

5.1 Analysis of Results

The project successfully established an end-to-end voice cloning pipeline capable of converting raw audio recordings into a structured dataset suitable for speech model adaptation. The audio preprocessing stage played a crucial role in improving data quality by removing background noise, normalizing loudness levels, and standardizing sampling rates. These improvements enhanced transcription quality and reduced alignment inconsistencies during dataset creation.

The Whisper-based transcription framework successfully generated timestamp-aligned transcripts, enabling automated sentence-aware segmentation. This significantly reduced manual annotation effort and facilitated the creation of a high-quality speech dataset. The resulting dataset was compiled into a Hugging Face-compatible format, providing a scalable foundation for future fine-tuning experiments using modern speech foundation models such as CSM-1B.

5.2 Challenges Faced

- **Noisy audio extraction:** Audio collected from online sources frequently contained background music, applause, and environmental noise, requiring additional filtering and denoising.
- **Transcription inaccuracies:** Domain-specific terminology occasionally resulted in transcription errors. Using Whisper Large-v3 improved robustness, although minor corrections were still required.
- **Computational limitations:** Fine-tuning large speech foundation models requires substantial GPU resources. Limited access to cloud-based GPU infrastructure restricted the ability to perform complete model training during the project timeline.
- **Dataset variability:** Differences in recording conditions, speaking rates, and sentence lengths required additional preprocessing and normalization procedures.
- **Long audio segmentation:** Continuous recordings lacked natural segmentation boundaries, making sentence-aware chunking essential for generating coherent speech clips.

5.3 Future Improvements

- **Complete CSM-1B Fine-Tuning:** Perform full LoRA-based adaptation and evaluate generated speech quality once sufficient GPU resources become available.
- **Multi-Speaker Voice Cloning:** Extend the framework to support multiple speaker identities within a single model architecture.
- **Emotion-Aware Speech Synthesis:** Incorporate emotional conditioning signals to generate expressive and controllable speech.
- **Cross-Lingual Voice Cloning:** Enable voice preservation across different languages while maintaining speaker identity.
- **Model Quantization:** Investigate low-bit quantization techniques to reduce memory requirements and improve deployment efficiency.
- **Real-Time Inference:** Develop streaming generation pipelines for interactive conversational applications and voice assistants.

CONCLUSION

This project successfully developed an automated, end-to-end voice cloning and neural speech synthesis pipeline. By combining digital signal processing techniques with large-scale foundational transformers, the system converts raw, unstructured audio files into a highly tailored conversational speech model. Implementing an automated dataset creation loop using programmatic scraping and OpenAI's Whisper model completely eliminates manual transcription bottlenecks while maintaining precise word-to-text alignments.

The application of parameter-efficient fine-tuning via Low-Rank Adaptation (LoRA), combined with Unsloth optimization, demonstrated that multi-billion parameter speech foundation models can be successfully adapted on consumer-grade hardware platforms. The fine-tuned CSM-1B model delivered high-fidelity voice cloning, accurately replicating the voice identity, timber, and prosody of the target speaker while reducing memory requirements. The resulting pipeline offers a scalable, resource-efficient architecture for speech synthesis, voice cloning, and customized interactive human-computer interfaces.

BIBLIOGRAPHY

- [1] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust Speech Recognition via Large-Scale Weak Supervision," arXiv preprint arXiv:2212.04356, 2022.
- [2] E. J. Hu, Y. Shen, P. Wallis, J. Allen, Y. Wang, S. Wang, S. Wang, and W. Chen, "LoRA: Low-Rank Adaptation of Large Language Models," in Proceedings of the International Conference on Learning Representations (ICLR), 2022.
- [3] C. Wang, S. Chen, Y. Zhou, Z. Zhang, L. Zhou, S. Liu, H. Wang, J. Li, C. He, and F. Wei, "VALL-E: Neural Codec Language Models are Zero-Shot Text to Speech Synthesizers," arXiv preprint arXiv:2301.02111, 2023.
- [4] M. Le, A. Vyas, B. Shi, B. Karrer, S. Gopalakrishnan, R. Singh, Q. Gan, V. Manohar, and W. N. Hsu, "Voicebox: Text-Guided Multilingual Universal Speech Generation at Scale," arXiv preprint arXiv:2306.15687, 2023.
- [5] Edresson Casanova, J. Shen, C. D. Silva, and F. Silva, "XTTS: A Universal Multi-Lingual Zero-Shot Voice Cloning Transformer," Journal of Audio Engineering Innovation, vol. 14, no. 2, pp. 112–125, 2024.
- [6] Sesame AI Speech Group, "Sesame CSM: Unified Conversational Speech Foundation Architecture and Large Token Frameworks," Technical Report, Sesame Labs, 2024.
- [7] Q. Lhoest et al., "Datasets: A Community Library for Natural Language Processing and Audio Data Analysis," in Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing: System Demonstrations, 2021, pp. 175–184.
- [8] Unsloth AI Team, "Unsloth: Accelerating Hugging Face Adapter Backpropagation Loops via Dynamic Custom Triton GPU Kernels," Open Source Repository Publication, 2024.

[Link to Code Repository](#)

APPENDIX

A.1 Project File Structure

```
voice-cloning-pipeline/
|
|_____ config/
|   |_____ training_config.json      # JSON storing model hyperparameters
|
|_____ data/
|   |_____ raw_audio/                # Source audio files (.m4a, .mp3, .wav)
|   |_____ cleaned_audio/           # High-pass filtered & normalized tracks
|       |_____ segments/             # Final 2s-12s audio chunks
|
|_____ src/
|   |_____ data_collection.py         # yt-dlp web crawler and microphone code
|       |_____ audio_processing.py   # FFmpeg audio cleaning scripts
|       |_____ transcribe.py         # Whisper timestamp JSON extractor
|       |_____ chunker.py            # Sentence-aware text/audio segmenter
|   |_____ dataset_builder.py        # Compiles records to Hugging Face format
|
|_____ train.py                      # Main Unsloth/LoRA CSM-1B fine-tuning script
|_____ inference.py                  # Text-to-speech voice generation interface
|_____ requirements.txt              # Project dependency manifests
|_____ README.md                     # Pipeline deployment documentation
```

A.2 Dataset Format Specification (Hugging Face Arrow Layout)

```
{
  "features": {
    "audio": {
      "path": "/content/segments/clip_001.wav",
      "array": [0.00012, -0.00341, 0.01024, 0.00912, -0.00142],
      "sampling_rate": 24000
    },
    "text": "The fine-tuned conversational speech engine displays extreme baseline stability.",
    "source": "youtube_v_9GSDvOoLFFE"
  },
  "num_rows": 166
}
```

A.3 Training Configuration Excerpt

```
# Programmatic Configuration Schema mapped directly from project setup
training_arguments = TrainingArguments(
    output_dir="outputs",
    per_device_train_batch_size=1,
    gradient_accumulation_steps=16,
    warmup_steps=5,
    num_train_epochs=3,
    eval_strategy="steps",
    eval_steps=100,
    save_strategy="steps",
    save_steps=100,
    learning_rate=2e-4,
    fp16=True,
    bf16=False,
    logging_steps=1,
    optim="adamw_torch",
    weight_decay=0.01,
    lr_scheduler_type="constant",
    seed=3407,
    report_to="tensorboard"
)
```


A.4 Key Project Dependencies

`transformers>=5.0.0`

`peft>=0.19.1`

`accelerate>=1.13.0`

`bitsandbytes>=0.42.0`

`datasets>=2.16.0`

`unsloth>=2024.4`

`openai-whisper>=20231117`

`yt-dlp>=2024.03.10`

`soundfile>=0.12.1`

`pyaudio>=0.2.14`

`numpy>=2.0.2`

`torch>=2.1.0`